



ENGIN 604 INTRODUCTION TO PYTHON FOR FINANCE APPLICATION: CAPITAL ASSET PRICING MODEL

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Developed by William Sharpe (1964), the CAPM (*Capital Asset Pricing Model*) starts from the idea that every investment carries two types of risk: systematic (market risk that cannot be diversified away) and unsystematic (the component of a stock's return that is not correlated with market movements). portfolio diversification can remove unsystematic risk; however, this is not the case for systematic risk.

The CAPM is a way of measuring systematic risk. William Sharpe finds that the return of a stock or portfolio is:

$$R_{i,t} = R_f + \beta_i(R_{m,t} - R_f)$$

Where:

- $R_{i,t}$: Expected return of the portfolio or stock.
- R_f : Risk-free rate.
- $R_{m,t}$: Expected market return.
- $R_{m,t} - R_f$: *Equity market premium*
- β_i : The CAPM beta.

According to the CAPM, β is the only relevant measure of risk, defined as:

$$\beta_i = \frac{\text{cov}(R_{p,t}, R_{m,t})}{\sigma_m^2}$$

If β is taken as the second coefficient of a simple OLS linear regression:

$$R_{i,t} - R_f = \alpha_i + \beta_i(R_{m,t} - R_f) + \varepsilon_{i,t}$$

Although the model above is not the CAPM, the regression allows the CAPM β to be estimated. In fact, if $\alpha_i = 0$ and $\text{Cov}(R_{m,t}, \varepsilon_{i,t}) = 0$ then:

$$\begin{aligned} R_{i,t} - R_f &= \beta_i(R_{m,t} - R_f) \\ R_{i,t} &= R_f + \beta_i(R_{m,t} - R_f) \end{aligned}$$

The CAPM equation is obtained.

1 Application

1. Import the files `stocks.pkl`, `sp500.pkl` and `rfree.pkl`.
 - `stocks.pkl`: indices for Facebook, Apple, Amazon, Netflix and Google from 2014:12 to 2020:12.
 - `sp500.pkl`: Closing price of the S&P 500 index from 2014:12 to 2020:12.
 - `rfree.pkl`: Risk-free rate from 2015:01 to 2020:12.
2. Write a function that computes the arithmetic return, defined as:

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}} \quad i = 1, \dots, 5 \quad t = 1, \dots, 73$$

3. Apply the function created in (2) to each element (closing prices) stored in the dictionary.
4. Apply the function created in (2) to the S&P 500 index. Then subtract the risk-free rate.

$$R_{m,t} = R_{sp500} - R_f$$

5. Write a function that weights each return by a fixed weight. The weights must sum to 1 and the *output* must have the risk-free rate R_f subtracted.

$$R_{p,t} = \mathbf{R} \cdot \mathbf{W} = \begin{pmatrix} r_{11} & r_{12} & \dots & r_{1i} \\ r_{21} & r_{22} & \dots & r_{2i} \\ \vdots & \vdots & \vdots & \vdots \\ r_{t1} & r_{t2} & \dots & r_{ti} \end{pmatrix}_{t \times i} \cdot \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_i \end{pmatrix}_{i \times 1}$$

Test your function with the following weights:

Table 1: Index & weights

Index	Weights
AAPL	0.25
AMZN	0.25
FB	0.20
GOOG	0.20
NFLX	0.10

6. Write a function that obtains the $\hat{\beta}$ of a linear regression using its matrix form:

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y})$$

7. Show that the previous result is equal to:

$$\hat{\beta} = \frac{cov(R_{p,t}, R_{m,t})}{\sigma_m^2}$$

8. Verify the result obtained in (6) using the `statsmodels` library.